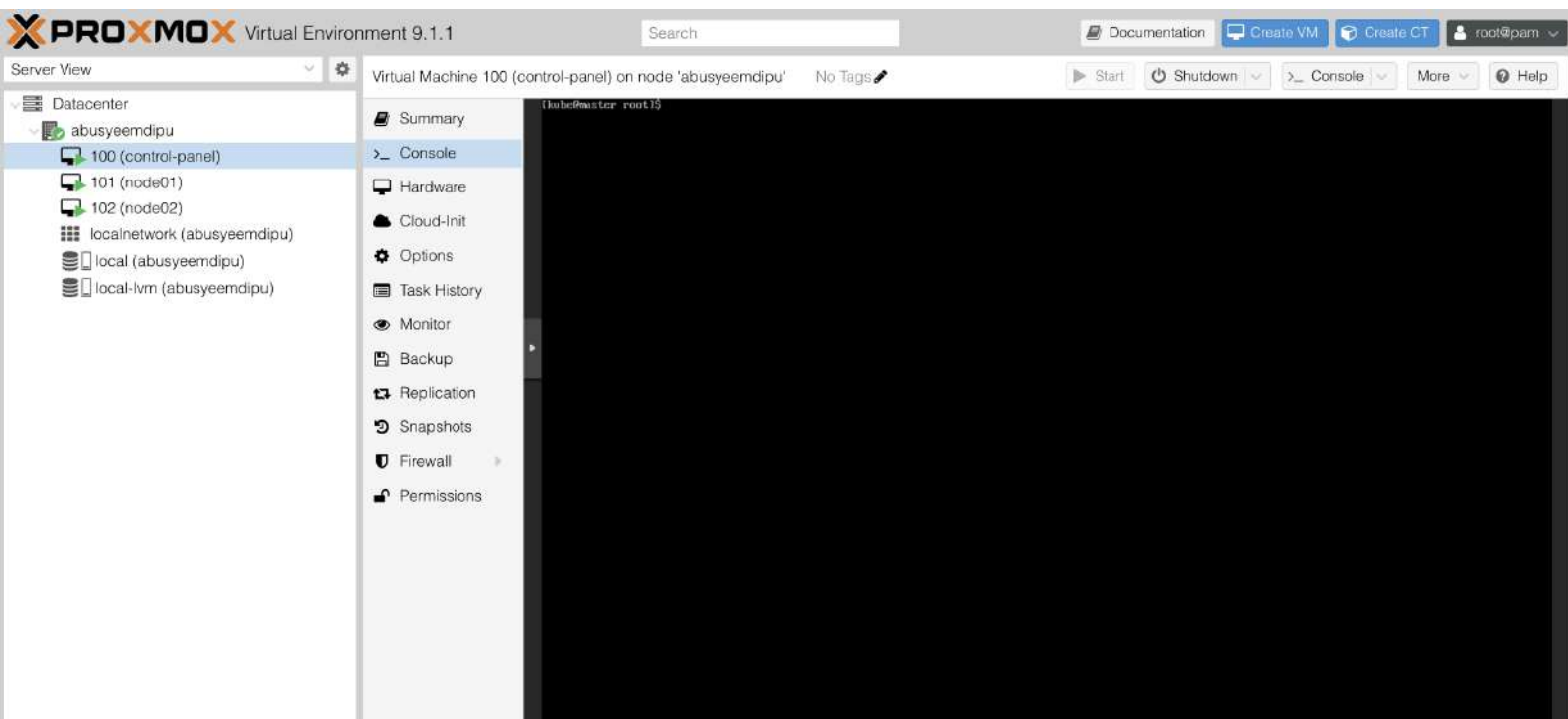




```
[kube@master root]$
[kube@master root]$
[kube@master root]$
[kube@master root]$ kubectl get nodes -o wide
NAME          STATUS    ROLES    AGE   VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE             KERNEL-VERSION        CONTAINER-RUNTIME
master        Ready     control-plane   10h   v1.28.15   192.168.178.49 <none>        Rocky Linux 9.7 (Blue Onyx) 5.14.0-611.5.1.el9_7.x86_64 containerd://2.2.1
node01       Ready     <none>      9h   v1.28.15   192.168.178.51 <none>        Rocky Linux 9.7 (Blue Onyx) 5.14.0-611.5.1.el9_7.x86_64 containerd://2.2.1
node02       Ready     <none>      9h   v1.28.15   192.168.178.52 <none>        Rocky Linux 9.7 (Blue Onyx) 5.14.0-611.5.1.el9_7.x86_64 containerd://2.2.1
[kube@master root]$
[kube@master root]$
[kube@master root]$
```

Kubernetes Cluster Setup – 1 Master, 2 Workers

Nodes: master (192.168.178.49), node01 (192.168.178.51), node02 (192.168.178.52)

Prerequisites

- Rocky/CentOS/RHEL Linux
- Internet access
- Root privileges on all nodes
- Hostnames configured and resolvable

Disable Swap & SELinux

```
swapoff -a
sed -i.bak 's/swap/s/ s/^/#/' /etc/fstab
setenforce 0
sed -i 's/^SELINUX=.*/SELINUX=permissive/' /etc/selinux/config
```

Kernel Modules & Sysctl

```
echo -e "overlay\nbr_netfilter" > /etc/modules-load.d/k8s.conf
modprobe overlay
modprobe br_netfilter
```

```
cat < /etc/sysctl.d/99-kubernetes.conf
net.bridge.bridge-nf-call-iptables=1
net.bridge.bridge-nf-call-ip6tables=1
net.ipv4.ip_forward=1
EOF
```

```
sysctl --system
```

Install containerd

```
dnf install -y yum-utils
dnf config-manager --add-repo https://download.docker.com/linux/centos/docker-ce.repo
dnf install -y containerd.io
containerd config default > /etc/containerd/config.toml
sed -i 's/SystemdCgroup = false/SystemdCgroup = true/' /etc/containerd/config.toml
systemctl enable --now containerd
```

Install Kubernetes Components

```
cat < /etc/yum.repos.d/kubernetes.repo
[kubernetes]
name=Kubernetes
baseurl=https://pkgs.k8s.io/core:/stable:/v1.28/rpm/
enabled=1
gpgcheck=1
gpgkey=https://pkgs.k8s.io/core:/stable:/v1.28/rpm/repodata/repomd.xml.key
exclude=kubelet kubeadm kubectl cri-tools kubernetes-cni
EOF
```

```
dnf install -y kubelet kubeadm kubectl --disableexcludes=kubernetes
systemctl enable kubelet
```

Initialize Master

```
kubeadm init --control-plane-endpoint=master
mkdir -p $HOME/.kube
cp /etc/kubernetes/admin.conf $HOME/.kube/config
chown $(id -u):$(id -g) $HOME/.kube/config
```

Install Calico

```
kubectl apply -f
https://raw.githubusercontent.com/projectcalico/calico/v3.27.3/manifests/calico.yaml
```

Join Workers

```
kubeadm token create --print-join-command
Run the output command on node01 and node02.
```

Validation

```
kubectl get nodes
kubectl get pods -A
```

LinkedIn Portfolio Tip

Share screenshots of `kubectl get nodes` and `kubectl get pods -A` with a short story of what you built.